

AKDENİZ UNIVERSITY
DEPARTMENT OF COMPUTER ENGINEERING

CSE321 Introduction to Probability Theory
Midterm Exam

Group A

- You have 45 minutes to solve 25 questions.
- Each question is +4 points. Each incorrect answer costs -1 points.
- Write your all answers to the answer sheet provided.

ID _____ Name _____

- 1) In a computer science experiment, the execution times (in milliseconds) of an algorithm are recorded as follows: 10, 12, 15, 20, 100. Which measure would be most appropriate to represent the central tendency of this data set?
A) Median
B) Geometric Mean
C) Mean
D) Harmonic Mean
E) Mode

- 2) A system has two components, A and B. The probability that component A fails is 0.1, and the probability that component B fails is 0.2. The system fails if either component fails. If the failures of A and B are independent, what is the probability that the system operates successfully?
A) 0.8 B) 0.72 C) 0.9 D) 0.98 E) 0.7

- 3) A sample of 10 software modules has defect counts with a sample variance of 25. If each defect count is increased by 5 (due to reclassification of minor issues as defects), what is the new sample variance?
A) 30
B) 50
C) 25
D) 75
E) Cannot be determined without data.

- 4) A data set consists of 100 observations with a mean of 50 and a variance of 25. If all observations are increased by 10% and then 5 units are added to each, what is the new mean and variance?
A) Mean = 60, Variance = 27.5
B) Mean = 55, Variance = 25
C) Mean = 60, Variance = 30.25
D) Mean = 60, Variance = 25
E) Mean = 60, Variance = 32.25

- 5) Which of the following graphs is best suited to display the relationship between two quantitative variables?
A) Histogram B) Scatterplot C) Pie Chart D) Boxplot E) Bar Chart

- 6) A data set has a mean of 20 and a standard deviation of 5. Assuming a bell-shaped distribution, approximately what percentage of data falls between 15 and 25?
A) 95% B) 75% C) 99.7% D) 68% E) 50%

- 7) If the z-score of a data point is -2, and the mean and standard deviation of the data set are 50 and 5 respectively, what is the value of the data point?
- A) 35 B) 50 C) 40 D) 55 E) 45

- 8) Given a grouped frequency distribution with class intervals and frequencies as below, estimate the mean:

Class Interval	Frequency
0 - 10	2
10 - 20	5
20 - 30	9
30 - 40	4

- A) 24
B) 22.5
C) 20
D) 18.5
E) Cannot be calculated without raw data
- 9) Which sampling method involves dividing the population into subgroups and then taking a simple random sample from each subgroup?
- A) Convenience Sampling
B) Stratified Sampling
C) Systematic Sampling
D) Simple Random Sampling
E) Cluster Sampling
- 10) Which measure of central tendency can be used for nominal (categorical) data?
- A) Range
B) Mean
C) Median
D) Standard Deviation
E) Mode
- 11) In a left-skewed distribution, which of the following is true?
- A) Median > Mean > Mode
B) Mode > Median > Mean
C) Mean > Median > Mode
D) Mean = Median = Mode
E) Mean < Median < Mode
- 12) A group of 12 people is traveling in three vehicles. If 4 people ride in each vehicle, in how many different ways can the people be assigned to the vehicles?
- A) 34,650 B) 11,550 C) 495 D) 11,880 E) 1,400
- 13) A computer science researcher collects data on the number of bugs found in 50 different software programs. The data is a type of:
- A) Qualitative continuous data
B) Qualitative discrete data
C) Quantitative continuous data
D) Quantitative discrete data
E) None of them.

- 14) An economy pack of highlighters contains 12 yellow, 6 blue, 4 green, and 3 orange highlighters. An experiment consists of randomly selecting one of the highlighters and recording its color. Find the probability that a blue or yellow highlighter is selected given that a yellow highlighter is selected.
- A) $\frac{2}{3}$ B) 0 C) $\frac{1}{2}$ D) $\frac{1}{3}$ E) 1
- 15) A study in the state of Georgia was conducted to determine the percentage of all community college students who have taken at least one online class. 1500 community college students were contacted and asked if they had taken at least one online class during their time at their community college. These responses were then used to estimate the percentage of all community college students who have taken at least one online class. What type of variable is being collected?
- A) quantitative data B) qualitative data
- 16) When constructing a histogram, which of the following statements is true?
- A) The width of each class should be equal
 B) The number of classes is arbitrary and does not affect the shape
 C) Classes should overlap to include all data points
 D) Histograms are used for qualitative data
 E) Frequency polygons cannot be derived from histograms
- 17) In a class of 30 students, 18 study Machine Learning (ML), 15 study Artificial Intelligence (AI), and 10 study both. If a student is selected at random, what is the probability that the student studies either ML or AI?
- A) 7/10 B) 1/2 C) 11/15 D) 23/30 E) 4/5
- 18) Which of the following is a measure of relative standing?
- A) Mean
 B) Interquartile Range
 C) Variance
 D) Z-score
 E) Standard Deviation
- 19) A boxplot is useful for displaying:
- A) The distribution of a qualitative variable
 B) Changes over time
 C) The relationship between two categorical variables
 D) The five-number summary of a data set
 E) The frequency of each data point
- 20) A number between 1 and 10, inclusive, is randomly chosen. Events A , B , C , and D are defined as follows.
- A : {The number is even}
 B : {The number is less than 7}
 C : {The number is less than or equal to 7}
 D : {The number is 5}
- Identify one pair of independent events.
- A) B and D B) A and B C) A and C D) B and C E) A and D

- 21) In a manufacturing process, the probability that a product is defective is 0.02. If 5 products are selected at random, what is the probability that at least one is defective?
- A) 9.6% B) 6.8% C) 2% D) 7.2% E) 8.7%
- 22) If all observations in a data set are multiplied by a constant 3, what happens to the standard deviation?
- A) It is multiplied by square root of 3
 B) It is divided by 3
 C) It is multiplied by 3
 D) It is multiplied by 9
 E) It remains the same
- 23) In a programming team of 5 developers, the probability that a developer knows Python is 0.6, and the probability that a developer knows Java is 0.8. Assuming the probabilities are independent, what is the probability that a randomly selected developer knows both Python and Java?
- A) 0.14 B) 0.48 C) 0.75 D) 0.68 E) 0.24
- 24) In a sample of size n , the mean is 40, and the standard deviation is 0. If another data point of value 40 is added to the sample, what are the new mean and standard deviation?
- A) Mean = 40, Standard Deviation > 0
 B) Mean = 40, Standard Deviation cannot be determined
 C) Mean > 40, Standard Deviation = 0
 D) Mean = 40, Standard Deviation = 0
 E) Mean and Standard Deviation increase
- 25) I. It can be negative
 II. It is measured in the same units as the mean
 III. It is less affected by outliers than the mean
 IV: It is the square root of the variance
 Which of the above statements are true regarding the standard deviation?
- A) I, II, and IV B) III and IV C) II, III, and IV D) II and IV E) I and II